

Li Ding

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🌐 <https://liding.info>

SUMMARY I work on AI control at Google DeepMind, focused on oversight and threat mitigation for increasingly capable agents. Previously, I worked on pretraining and efficiency for the Gemini on-device models. I am interested in scalable, exploration-driven methods for safe, reliable agents in open-ended environments.

EDUCATION

University of Massachusetts Amherst
Ph.D. in Computer Science 2020 - 2024

- Committee: Lee Spector (Chair), Scott Niekum, Subhansu Maji, Jeff Clune
- Dissertation: "Optimization with intrinsic diversity: Towards generalizable, safe, and open-ended learning"

Massachusetts Institute of Technology
Graduate Study in EECS (non-degree) 2019 - 2020

University of Rochester
M.S. in Data Science 2016 - 2017

WORK EXPERIENCE

Google DeepMind
Research Engineer 2026.1 - present

- Built an asynchronous monitoring system that classifies agent misbehavior against a threat taxonomy and escalates for human review, analyzed 1M+ coding agent trajectories.
- Developed live monitoring systems for agents (Antigravity, Gemini Spark) to detect and mitigate misbehavior such as unauthorized data deletion and overeagerness.
- Contributed to a defense-in-depth AI control and safety framework that treats capable agents as potential insider threats, scaling oversight and response to model capability.

Google
Research Engineer 2024.7 - 2026.1

- Worked on pretraining and model efficiency for Gemini on-device models (Gemini Nano 3, Gemma 4), including architecture, quantization, and efficient inference on TPU/NPU.
- Research on efficient and privacy-preserving LLM adaptation methods for personalization.
- Developed Gemma models in PyTorch, integrated with popular open-source libraries.

Massachusetts Institute of Technology
Research Engineer 2017.10 - 2020.6

- Research on deep learning for video scene segmentation, developed MIT DriveSeg dataset.
- Developed a multitask framework for eye dynamics and cognitive load assessment.

SELECTED PUBLICATIONS

[GOOGLE SCHOLAR](#)

Quality Diversity through Human Feedback: Towards Open-Ended Diversity-Driven Optimization

Li Ding, Jenny Zhang, Jeff Clune, Lee Spector, Joel Lehman **ICML 2024**

- QDHF learns diversity metrics from human feedback and optimizes exploration of novel solutions, enhancing task-solving of RL agents and creativity of generative models.

Pareto-Optimal Learning from Preferences with Hidden Context

Ryan Bahlous-Boldi, Li Ding, Lee Spector, Scott Niekum

RLC 2025

- POPL learns Pareto-optimal policies/rewards in RLHF, catering diverse group preferences without needing group labels, thus offers safe and fair alignment of RL agents and LLMs.

Probabilistic Lexicase Selection

Li Ding, Edward Pantridge, Lee Spector

GECCO 2023

Optimizing Neural Networks with Gradient Lexicase Selection

Li Ding, Lee Spector

ICLR 2022

Value of Temporal Dynamics in Driving Scene Segmentation [MIT DriveSeg Dataset]

Li Ding, Jack Terwilliger, Rini Sherony, Bryan Reimer, Lex Fridman

Trans. IV 2021

Weakly-Supervised Action Segmentation with Iterative Soft Boundary Assignment

Li Ding, Chenliang Xu

CVPR 2018

INTERNSHIPS & COLLABS

Google Research

Research Intern

2023

- Host: Masrour Zoghi. Developed a JAX optimizer for efficient neural architecture search. Paper published in NeurIPS 2023 workshops.

CarperAI (Stability AI Research Lab)

Research Collaborator

2023

- Host: Joel Lehman. Developed an RLHF algorithm to enhance the creativity of generative models. Paper published in ICML 2024.

Meta

Research Scientist Intern

2022

- Hosts: Wenliang Zhao & Hang Zhang. Worked on efficient vision transformers architectures and transfer learning methods.

TEACHING

MIT 6.S094: Deep Learning for Self-Driving Cars (Teaching Assistant)

2018 - 2019

MIT Robocar Workshop (Co-instructor w/ Tom Bertalan)

2018

SERVICES

Reviewer for ICLR, NeurIPS, ICML, JMLR, CVPR, ICCV, ECCV, etc.

2020 - present

Ph.D. Admissions Committee (UMass Amherst CICS).

2024

OPEN SOURCE PROJECTS

[google-research/ev3](#): a meta-learning and architecture search optimizer in JAX.

- Main contributor for implementation and maintenance.

[pyribs](#): an open-source library for diversity-driven optimization.

- Contributed the code, demo, and tutorial for QDHF.

[mit-deep-learning](#): MIT Deep Learning Open Courses (10k+ stars).

- Main contributor for tutorials and coding assignments.

SKILLS

Programming: Python, C++, Go, PyTorch, JAX, TensorFlow.

LLMs & ML: Pretraining, fine-tuning (SFT, PEFT), alignment (RLHF / preference optimization), quantization-aware training (QAT), inference optimization.

AI Safety: AI control, agent monitoring, scalable oversight, (coding) agent simulation / sand-boxing, behavioral/anomaly detection, threat modeling.